

School of Computer Science Engineering and Technology

Lab No. - 24

Course-B. Tech.	Type- Core
Course Code- CSET301	Course Name- Artificial Intelligence and Machine Learning
Year- 2025	Semester- Odd
Date-	Batch- 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
Q1					√	

Lab – Implementation of RNN (Recurrent Neural Network)

Total Marks: 2

Objective:

To understand the implementation of Recurrent Neural Networks (RNNs) for text classification tasks. The experiment aims to demonstrate how RNNs learn and capture sequential dependencies in textual data and how this ability makes them more effective than traditional Artificial Neural Networks (ANNs) or other machine learning models when dealing with data that has temporal or contextual relationships.

Problem Statement

Develop a sentiment analysis model using RNN architecture to classify Amazon product reviews as positive or negative. Experiment with different hyperparameters to evaluate their impact on performance metrics like accuracy and loss. Additionally, compare the performance of the RNN model with a baseline Artificial Neural Network (ANN) model to analyze how sequence learning enhances text classification accuracy.

In this task, you will:

- Perform data preprocessing by cleaning and tokenizing the text.
- Use word embeddings to represent text data in dense vector space, capturing semantic relationships between words.

- Implement a Recurrent Neural Network (RNN) model for sentiment classification, experimenting with different hyperparameters such as embedding size, hidden units, dropout, optimizer, and number of epochs, such as listed below:
 - SIZE OF THE EMBEDDING LAYER (embedding_size): 16, 32
 - SIZE OF THE HIDDEN LAYER (hidden_size): 64, 128, 256
 - DROPOUT (dropout): 0, 0.3, 0.5
 - BATCH SIZE (batch_size): 32, 64, 128, 256
 - NUMBER OF EPOCHS (num_epochs): 10, 15, 20, 25
 - OPTIMIZER (optimizer): 'adam', 'rmsprop'
- Train the model to classify reviews as positive or negative, and monitor its learning progress using training and validation accuracy and loss curves.
- Compare the RNN model's performance with a baseline Artificial Neural Network (ANN) to highlight the benefits of sequence learning in text-based tasks.
- Visualize model performance through graphs (accuracy/loss curves, confusion matrix) and interpret results to understand the effectiveness and limitations of RNNs in handling sequential data for sentiment analysis.

Download the dataset from:

<https://www.kaggle.com/datasets/vasserh/amazon-product-reviews-dataset/data>

About Dataset:

The Amazon Product Reviews dataset consists of detailed product and review information, including attributes such as product ID, brand, categories, pricing, manufacturer details, and customer feedback. It contains both product metadata and user-generated content such as review text, ratings, recommendations, and review dates. **For this lab, only the 'reviews.text' and 'reviews.rating' columns are used to analyze customer sentiment through text classification.**

Questions:

1. Data Preprocessing

- a) Load the Amazon Reviews dataset and display:
 - Total number of samples.
 - Missing values per column.
- b) Clean and preprocess the text data by performing:
 - Lowercasing of text.
 - Removal of URLs, numbers, punctuation, and special characters.
 - Removal of stopwords.
 - Lemmatization of tokens.
- c) Encode the star ratings into three sentiment categories:

- 1–2 → Negative
- 3 → Neutral
- 4–5 → Positive

d) Convert the cleaned text into numerical sequences using TextVectorization, and fix the sequence length through the parameter `output_sequence_length`.

e) Split the dataset into training and testing sets.

2. Model Implementation

a) Build a Recurrent Neural Network (RNN) model using the Keras Sequential API with the following layers:

- Embedding layer
- SimpleRNN layer
- Dropout layer
- Dense output layer with softmax activation

b) Compile the model using:

- Loss function: `categorical_crossentropy`
- Optimizer: `adam`
- Metric: `accuracy`

c) Train the model for a given number of epochs and observe training and validation accuracy.

3. Result Evaluation and Visualization

a) Plot training and validation accuracy across epochs.

b) Evaluate the model on the test data and report Validation accuracy and Validation loss

c) Generate a confusion matrix and a classification report to assess the model's performance

4. Comparison with Baseline Model

a) Implement a baseline Artificial Neural Network (ANN) using the same training and test data.

b) Compare the accuracy and loss curves of ANN and RNN models.

c) Discuss how the RNN model's ability to capture sequential word dependencies helps it perform better in text classification compared to the ANN.